Projects 1

Pre

CS 3244
Machine Learning

NUS National University of Singapore | Computing

Project Objectives

Learning outcomes: Higher order construction and interactive learning of ML.

- Producing output that contain ideas that go beyond the presented information
- Dialoguing substantively on the same topic, transacting on each others' contribution

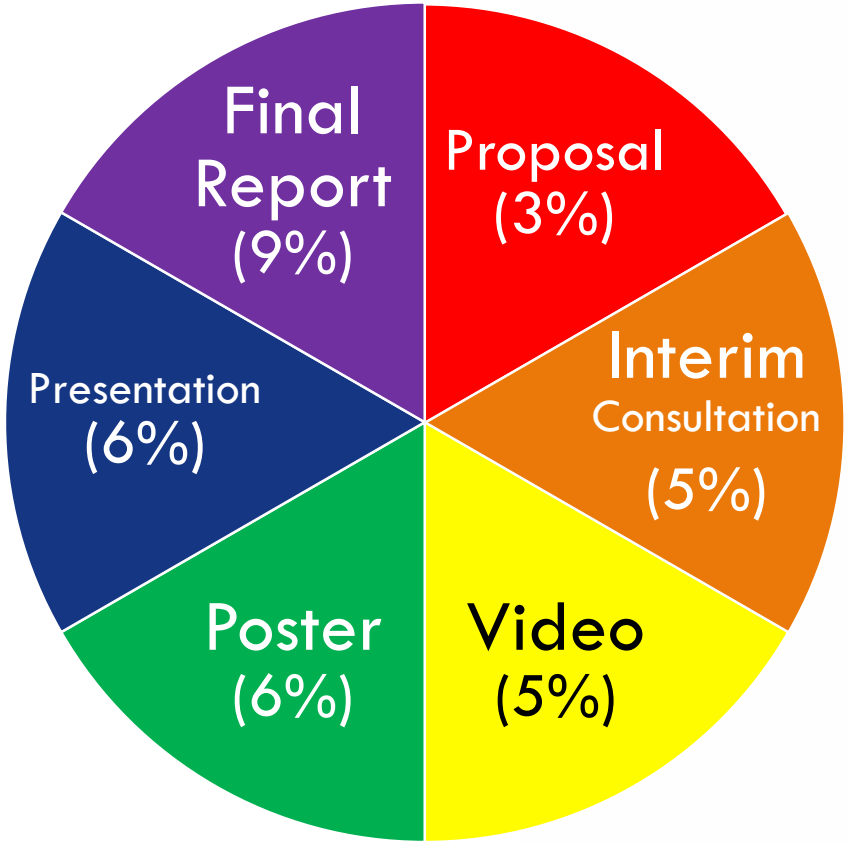
Active Verbs

- Reflecting
- (Constructive) Criticizing
- Comparing and Contrasting
- Creating
- Communicating
- Interacting among teams, intra-acting within team

Components

Start here:
http://bit.ly/cs3244-project-guidelines

And here:
http://bit.ly/cs3244-project-quick-start



Each project component has its own already published guidelines and grading rubric.

1. http://bit.ly/cs3244-project-proposal
2. http://bit.ly/cs3244-interim-project-consultation
3. http://bit.ly/cs3244-poster-presentation
4. http://bit.ly/cs3244-project-final-report

Important Dates

Week 02 (19 Aug 23:59): Form subteams of 3 students

Week 04 (4 Sep 23:59): Project Proposals Due

Week 05 (11 Sep 23:59): Peer reviews for Project Proposal due

Week 07 or 08: Meet with staff for project consultation. Decide on your team's STePS status in conjunction with the staff.

Week 09 or 10: For STePS teams, meet staff for the 2nd project consultation.

Week 10 (23 Oct 23:59): Non-STePS teams: Posters and Videos Due

Week 11 (30 Nov 23:59): Non-STePS teams: Peer grading of the Project Posters and Videos Due

Week 12 (6 Nov 23:59): STePS teams: Posters and Videos Due

Week 13 (11 Nov 18:00-22:00): STePS teams to present at STePS

Week 13 (13 Nov 23:59): STePS teams: Peer grading of the Project Posters and Videos Due

Week 13 (13 Nov 23:59): All teams: Project Reports due.

Executing your project: Common Issues

Dataset scale and compute resources.

- Deep learning methods take time and resource availability. Don't underestimate this part. Test for scale.

Referencing prior work.

- Esp. Kaggle work: be forthright about whom you looked up.
- Describe how your work fits with the lecture materials.

Allowing <u>everyone</u> in the team to work on the ML portions.

- All members should be doing part of the ML modeling work, in some way (different models, regularization, features)

Crediting Others

Cite scientific publications where applicable.

Give source and credit to images that you source that are attributable. Elect not to use copyrighted images by ensuring they are open for public use.

Give links to website, blogs and references where applicable.

A Reference section on your poster may be appropriate; mandatory for your final report.

Poster Summary

Goal: Summarized version of your intended final report.

Requirement: your title, abstract and your team members' names (and affiliations if you wish)

All other aspects are up to you. Do make it easy to give the highlights of your work within a 3-4 minute verbal presentation.

See the final report guidelines for grading criteria: i.e., Originality, Significance, Relevance, Quality of Argument, Related Work, Clarity and Scope.

Poster – Formatting

A1 sized (594 x 841mm)

Portrait Orientation (requirement from STePS).

You can get one copy of your poster printed free of charge from the SoC Technical Services (COM1 #01-06). Please check that you include your project ID and STePS sponsor logos – See STePS FAQ.

Use a high resolution .pdf or .pptx file (at least 300dpi), for best results.

Submit the link to an image (png, jpg, pdf) version of your poster to both isteps.comp.nus.edu.sg (for public consumption) and Coursemology (for peer review).

Video Summary

Goal: Science communication for the layman audience.

3 minute (strict) or shorter video summary of your teams' work, similar to other short ML videos found on the web.

To communicate the high level vision of your project and passion for your project's motivation and findings.

Technical details can be omitted. Keep your video of interest to the general audience, and independent of mentioning CS3244.

To help your own recruitment for internship or full-time job employment.

Finding a Project

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

4. Clarity (how clearly are goals and achievements presented)
5. Scope/Size (in proportion to size of group)
6. Significance (are the questions you are asking interesting)

Project Showcase

Below is a listing of the projects from this past semester. We have marked out some projects as 🏆 'notable' 🏆 - these projects either exhibit a sufficient degree of rigor or the team has shown considerable maturity over the duration of the course.

- 🏆 **Notable** *Generating word embeddings for Singlish to be used in sentiment analysis*

By Lee Ming Liang, Sheryl Toh , Perry Wang Zhi Ming, Chew Chia Sin, Xiao Yunhan, Le Trung Hieu

"Hi, how are you?" this sentence may be simple for us to understand yet is incomprehensible for a computer. Our project aims to explore how can we model the daily language of Singapore (Singlish) using mathematical representation. One common way to represent words would be the use of vectors to capture the semantic meaning of the words. Another name such vector representation is known as the word embedding.

Since this is an unsupervised learning task, there is no practical intrinsic evaluation on how well the generated embedding can capture the semantic and syntactic meaning of the words. Therefore, we will be performing an extrinsic evaluation on the embeddings generated. We will be applying the embedding generated in sentiment analysis on sentences to see how well the word embeddings capture the semantic meaning of the Singlish words. We found that the Continuous Bag Of Words (CBOW) Word2Vec model works the best, achieving the highest F1-score on the sentiment classification task.

- *Real-world Image Recognition for Multiple Human Attributes*

By Niu Yunpeng, Wang Debang, Yu Pei, Daniel Koh, Goh Wen Zhong, Tan Ying Lin

Ideating

Stanford's CS 229 Project Guidelines
http://cs229.stanford.edu/projects.html

CS229

CS229 Final Project Information

One of CS229's main goals is to prepare you to apply machine learning algorithms to real-world tasks, or to leave you well-qualified to start machine learning or AI research. The final project is intended to start you in these directions.

For group-specific questions regarding projects, please create a private post on Piazza, or email cs229-project@cs.stanford.edu. Please first have a look through the frequently asked questions.

Previous projects

2017	2016	2016 (Spring)	2015	2014
2013	2012	2011	2010	2009
2008	2007	2006	2005	2004

Project Topics

Your first task is to pick a project topic. If you're looking for project ideas, please come to project office hours, and we'd be happy to brainstorm and suggest some project ideas. In the meantime, here are some suggestions that might also help.

CS3244 Project Guidance

Talk with our many (Project) Teaching Assistants!

Graduate Teaching Assistants

HU Hengchang
e0409766@u.nus.edu

Joel LEE
joel_lee@u.nus.edu

LIANG Yuxuan
e0427783@u.nus.edu

Martin STROBEL
mstrobel@comp.nus.edu.sg

Abhinav Ramesh Kashyap
abhinav@comp.nus.edu.sg

Undergraduate Teaching Assistants

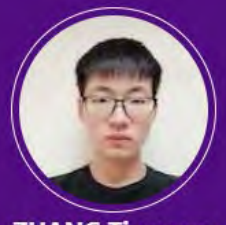
Shreyas KUTHANOOR PRAKASH
shreyas@u.nus.edu

LIU Yingnan
liu.yingnan@u.nus.edu

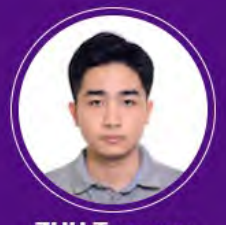
NI Tianzhen
nitianzhen@u.nus.edu

Marcus LEE
marcus_lee@u.nus.edu

LEOW Wen Bin
leowwb@u.nus.edu



ZHANG Tianyang
e0253753@u.nus.edu



ZHU Tongyao
tongyao.zhu@u.nus.edu

CS 229 (Continued)

- **Deep learning projects:** Since CS229 discusses many other concepts besides deep learning, we ask that if you decide to work on a deep learning project, please make sure that you use other material you learned in the class as well. For example, you might set up logistic regression and SVM baselines, or do some data analysis using the unsupervised methods covered in class. … Finally, training deep learning models can be very time consuming, so make sure you have the necessary compute.
- **Preprocessed datasets:** While we don't want you to have to spend much time collecting raw data, the process of inspecting and visualizing the data, trying out different types of preprocessing, and doing error analysis is often an important part of machine learning. Hence if you choose to use pre-prepared datasets (e.g. from Kaggle, the UCI machine learning repository, etc.), we encourage you to do some data exploration and analysis to get familiar with the problem.
- **Replicating results:** Replicating the results in a paper can be a good way to learn. However, we ask that instead of just replicating a paper, also try using the technique on another application, or do some analysis of how each component of the model contributes to final performance.

Project Resources

- Compute
- Video
- Datasets

- Kaggle
  - Datasets
  - Problems
  - Kernels

- Posters
  - Stanford CS 229 / 14th, 15th and 16th STePS

- pc_notebk_ipad
- personal_webpages
- printing
- research_support
- soc_student-helper_
- software
- solaris_zone
- ssh
- teaching_support
- unix
- virus_anti-virus
- web
- wfh
  - 2fa
  - address_book_setu
  - casual-loan-items
  - central_web_hostin
  - leaving_soc
  - motd
  - mysoc_intranet
  - soc_research_repos
  - soc_webdav_servic
  - visitor
  - windows_support
- tech
- faqs
- contact

xgpd0 - xgpd9	10	Ubuntu 16.04	2 x Xeon Silver 4108 (1.8Ghz)	128GB DDR4	960GB SSD & 6TB SATA HDD	10GbE	2x NVIDIA Titan V GPU	2018	Asus
xgpe0 - xgpe9	10	Ubuntu	2 x Xeon Silver 4116 (2.1Ghz)	256GB DDR4	960GB SSD & 3.84TB SSD	10GbE	1x NVIDIA Titan RTX GPU	2019	Asus
xgpe10	1	Ubuntu	2 x Xeon Silver 4116 (2.1Ghz)	256GB DDR4	960GB SSD & 3.84TB SSD	10GbE	2x NVIDIA Titan RTX GPU	2019	Asus
xgpe11	1	Ubuntu	2 x Xeon Silver 4116 (2.1Ghz)	512GB DDR4	960GB SSD & 3.84TB SSD	10GbE	2x NVIDIA Titan RTX GPU	2019	Asus
xgpf0 - xgpf9	10	Ubuntu	2 x Xeon Silver 4116 (2.1Ghz)	256GB DDR4	960GB SSD & 3.84TB SSD	10GbE	1x NVIDIA Tesla T4 GPU	2019	Asus
xgpf10	1	Ubuntu	2 x Xeon Silver 4116 (2.1Ghz)	256GB DDR4	960GB SSD & 3.84TB SSD	10GbE	2x NVIDIA Tesla T4 GPU	2019	Asus
xgpf11	1	Ubuntu	2 x Xeon Silver 4116 (2.1Ghz)	512GB DDR4	960GB SSD & 3.84TB SSD	10GbE	2x NVIDIA Tesla T4 GPU	2019	Asus
cgpa0 - cgpa3	4	Ubuntu	2x Xeon Silver 4214	128GB	960GB SSD & 8TB HDD	1GbE	4x NVIDIA RTX2080	2020	Asus
cgpb0	1	Ubuntu	2x Xeon Silver 4210	256GB	3.2TB SSD	10GbE	2x NVIDIA Tesla V100 GPU	2020	HPE

Amazon Credit

AWS sponsors our class for 10,000 USD providing free usage to support our course for students that participate.

We encourage those of you who already know how to use AWS or wish to learn it to contact Hengchang Hu (Project Coordinator).

All teams can take at least 200 USD of credit. We encourage you to start with a smaller VM first if you are unfamiliar with AWS.

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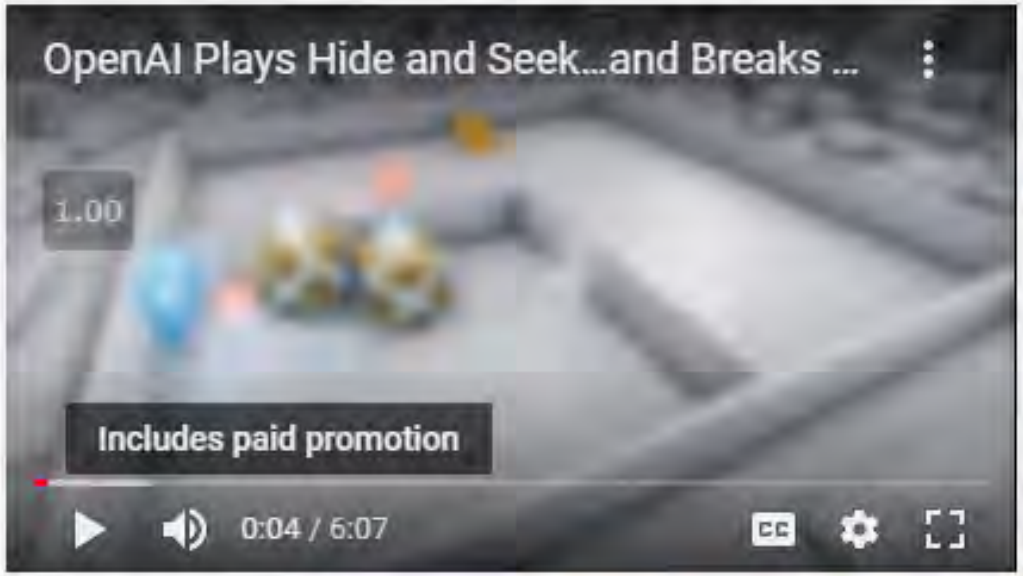

OpenAI Plays Hide and Seek...and Breaks The Game! 🤯

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Their blog post is available here:
https://www.wandb.com/articles/better...

📝 The paper "Emergent Tool Use from Multi-Agent Interaction"

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tasets found

Data from: The National University of Singapore SMS...
www.kaggle.com
zip
Updated Aug 7, 2017

SMS Spam Collection Dataset
www.kaggle.com
zip
Updated Dec 2, 2016

SMS Spam
data.world
zip

kaggle

Data from: The National University of Singapore SMS Corpus

A corpus of more than 67,000 SMS messages in Singapore English & Mandarin

Related Article

Explore at Kaggle

zip (70570817 bytes)

Dataset updated Aug 7, 2017

Authors

Rachael Tatman

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OSIC Pulmonary Fibrosis Progression
Predict lung function decline
Featured • 2 months to go • Code Competition • 675 Teams
$55,000

SIIM-ISIC Melanoma Classification
Identify melanoma in lesion images
Featured • 13 days to go • 3062 Teams
$30,000

Google Landmark Retrieval 2020
Given an image, can you find all of the same landmarks in a dataset?
Research • 13 days to go • Code Competition • 384 Teams
$25,000

Cornell Birdcall Identification
Build tools for bird population monitoring
Research • a month to go • Code Competition • 640 Teams
$25,000

Google Landmark Recognition 2020
Label famous (and not-so-famous) landmarks in images
Research • 2 months to go • Code Competition • 60 Teams
$25,000

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arXiv Dataset 58
Cornell University · Link
3 months · 2 GB · 8.8 · 3 Files (JSON)

Chipotle Locations 10
Jeffrey Braun
8 days · 124 KB · 10.0 · 2 Files (CSV, JSON) · 1 Task

Indian School Education Statistics 13
Vidya Pb
13 days · 24 KB · 10.0 · 7 Files (CSV) · 1 Task

ASHRAE Global Thermal Comfort Database II 6
Clayton
15 days · 3 MB · 10.0 · 1 File (CSV) · 2 Tasks

This American Life Podcast Dialog Transcripts 6
Shuyang Li
3 days · 76 MB · 10.0 · 6 Files (JSON, other) · 1 Task

Open Tasks

Contest #6: Lewis vs. Oleinik UF...
1 Submission · In Ultimate UFC Dataset

Order Brushing
1 Submission · In Shopee Code League 202...

Perform a consistent pattern of ...
0 Submissions · In COVID-19 SOLES Tables

Wind Power Forecasting
0 Submissions · In Wind Power Forecasting

How do human and personal fac...
0 Submissions · In ASHRAE Global Thermal ...

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70 Fraud Detection in Insurance Claims
3h ago in Insurance Claim gpu
Py 15

10 EfficientDet TPU
5h ago with multiple data sources tpu
Py 1

359 Melanoma. Pytorch starter. EfficientNet
12h ago with multiple data sources 0.925 gpu
Py 114

3 Basic visualization techniques
39m ago in Iris Species gpu
R 1

9 Getting to the top in house prices
5h ago in House Prices: Advanced Regression Techniques gpu
Py 0

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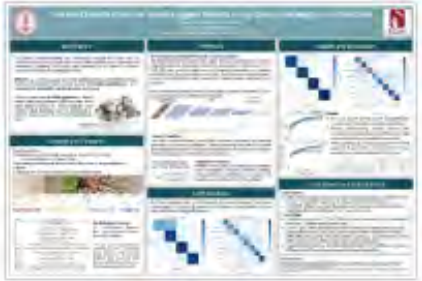
CS229

CS 229 projects, Spring 2020

All project posters and reports



Terrain Classification for Small Legged Robots Using Deep Learning on Tactile Data

General Machine Learning

Hojung Choi, Rachel Thomasson

[report][poster]

Application of machine learning methods to identify and categorize radio pulsar signal candidates

Physical Sciences

Serena Debesai, Carmen Gutierrez, Nazli Ugur Koyluoglu

[report]

Using Machine Learning Models to Predict S&P500 Price Level and Spread Direction

Finance & Commerce

Alex Fuster, Zhichao Zou

[report]

Exploring Adversarial Training for Out-of-

Project Proposal

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Proposal (3% final grade)

The project proposal should outline what you want to do in your project. We will give you feedback on your proposal to make sure the project you are proposing is feasible and appropriate. The proposal should contain the following sections:

Motivation. Explain why this project is interesting and important.

Statement of the Problem/Task. A statement of the problem, issue, or task that you're interested in studying. In particular, try to formulate the key questions (2 to 4 questions is probably a good number) that you will answer in the project.

General Approach. A high-level description of the general approach you'll use to address the questions. This should include how you will evaluate and what evidence you are planning to gather (e.g. how you can answer the questions through experiments on data).

Resources. A list of resources you have/need to conduct the project. This includes reading, software, datasets, etc. How are you planning to get these resources?

Schedule. A schedule of work indicating the dates by which you plan to complete components of the project. Make sure the schedule is plausible.

Predicting Watching Activity on CS3244 Video Analytics

Motivation

Explain why this project is interesting and important.

No more than 1 paragraph for the proposal. 1 general sentence that is grandma understandable. 2 sentences understandable by your classmates.

Statement of the Problem/Task

A statement of the problem, issue, or task that you're interested in studying. In particular, try to formulate the key questions (2 to 4 questions is probably a good number) that you will answer in the project.

As this proposal document is self-contained, you should restate your project topic and domain. The information you provided in the submission system is to help with appropriate review assignment.

Domain: Education
Approach: Deep Learning

Goal: 1-2 questions
Stretch Goal: 2 additional questions

General Approach

A high-level description of the general approach you'll use to address the questions. Sketch out what evidence you are planning to gather (e.g. how you can answer the questions through experiments on data).

We ask that if you decide to work on a deep learning project, please make sure that you use other material you learned in the class as well. time to obtain data for this project (1 week), so you need to carefully threshold the time (set a firm deadline) for when you cut off timing. You may need to run a pilot to test how long it takes you to do work.

Some datasets (esp. Kaggle) may be too large scale for you to work comfortably on. Decide how you are going to sample the dataset to make a reasonable decision.

Disclosure of prior work (especially for the interim report; as links to references; or public Github repos).

Schedule / Role Assignment

A schedule of work indicating the dates by which you plan to complete components of the project. Make sure the schedule is plausible.

You may find that a table format with the remaining weeks of the course helpful to describe this goal.

Generally we will have two work groups consisting of members from the two subgroups: Subgroup1 (Members 1-3) and Subgroup 2: (Members 4-6)

Member 1-4: SoC student
Member 5-6: Exchange Student
Member 3 will be away to represent NUS on competition on Week 6.

Week 5	- Scope defined and fleshed out, Consultations
Week 6	- Group 1 Software - Group 2 Literature Search - Interim Report

We ask that if you decide to work on a deep learning project, please make sure that you use other material you learned in the class as well. Deep Learning approaches need sufficient data, so ensure you have enough compute and time. Don't fail because your team has difficulty obtaining resources and time.

Evaluation

Include how you will evaluate your project. Propose what your team thinks is a satisfactory project outcome (C grade) and an excellent project outcome (A grade).

Satisfactory Outcome – Seek to finish a satisfactory project within the first XX% of your time and set stretch goals for your project. Underpromise and overdeliver.

Always have a defensible project in case human resources are unavailable for whatever reason. It's easier to extend a work than do damage control for an overly broad project.

Excellent Outcome – Good projects are close in quality to academic project outputs. See useful projects from other portals. Do active research on other's projects. Excellent projects must go beyond performance benchmarks and shed post ML analyses and relate to the course.

How do you build or extend work on the domain, dataset, metric or ML models?

Resources

A list of resources you have/need to conduct the project. This includes reading, software, datasets, etc. How are you planning to get these resources?

Back up plan: Work Plan A and Work Plan B (builds on WP A as a minimum). Scope the data that you plan to have. There's not a lot of

Week 6	- Group 1 Software - Group 2 Literature Search - Interim Report
Week 7 (30% less effort)	[Midterms] Implementation, v1
Week 8	Implementation, v2
Week 9	Experimentation
Week 10	Reporting: Captain: Member 1 - Member 2 – Video - Member 3 – Poster - Member 4-6 – Report
Week 11 (30% less effort)	Lag week

References

Read the other document *Project Guidelines* in full detail before writing your plan.

Consider your teammates availability and commitments and be clear about the notional effort that your members wish to put in.

10. Proposal Checklist *

Please check that your project proposal provides information to satisfy the reviewing criteria. Have you provided information to describe:

· Your project's motivation and significance?
· The general approach to the project?
· Your evaluation plan?
· Proposed grading criteria for both a satisfactory (C grade) and an excellent (A grade) project?
· Projected resources to be used?
· Roles of each member of the group and the project schedule?

☑ I agree

11. Submission Checklist *

Please certify that your proposal document is:

· Single A4 page in PDF format
· Conforms to the MS Word template provided
· Does not contain any student identiable information

☑ I agree

STePS

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

To STePS or not to STePS

Weeks 07-08:
Interim Project Consultations

Non-STePS

Week 10:
Project Video and Posters due

Week 11:
Presentation to Staff

STePS only

Weeks 09-10:
2nd Project Consultation

Week 12:
Project Video and Posters due

Week 13:
Public Presentation at STePS

Week 13: Final Report Due

STePS instructions

You'll need to update your title, abstract, poster image URL and video link to isteps.comp.nus.edu.sg for presentation to the public and peers on 11 Nov. There is a link to useful documentation to set up a profile.

In the 100 word abstract, include:

- A description of the problem;
- A description of what your project does to solve it;
- An indication of who might be interested in your project.

13 November 18:00-22:00

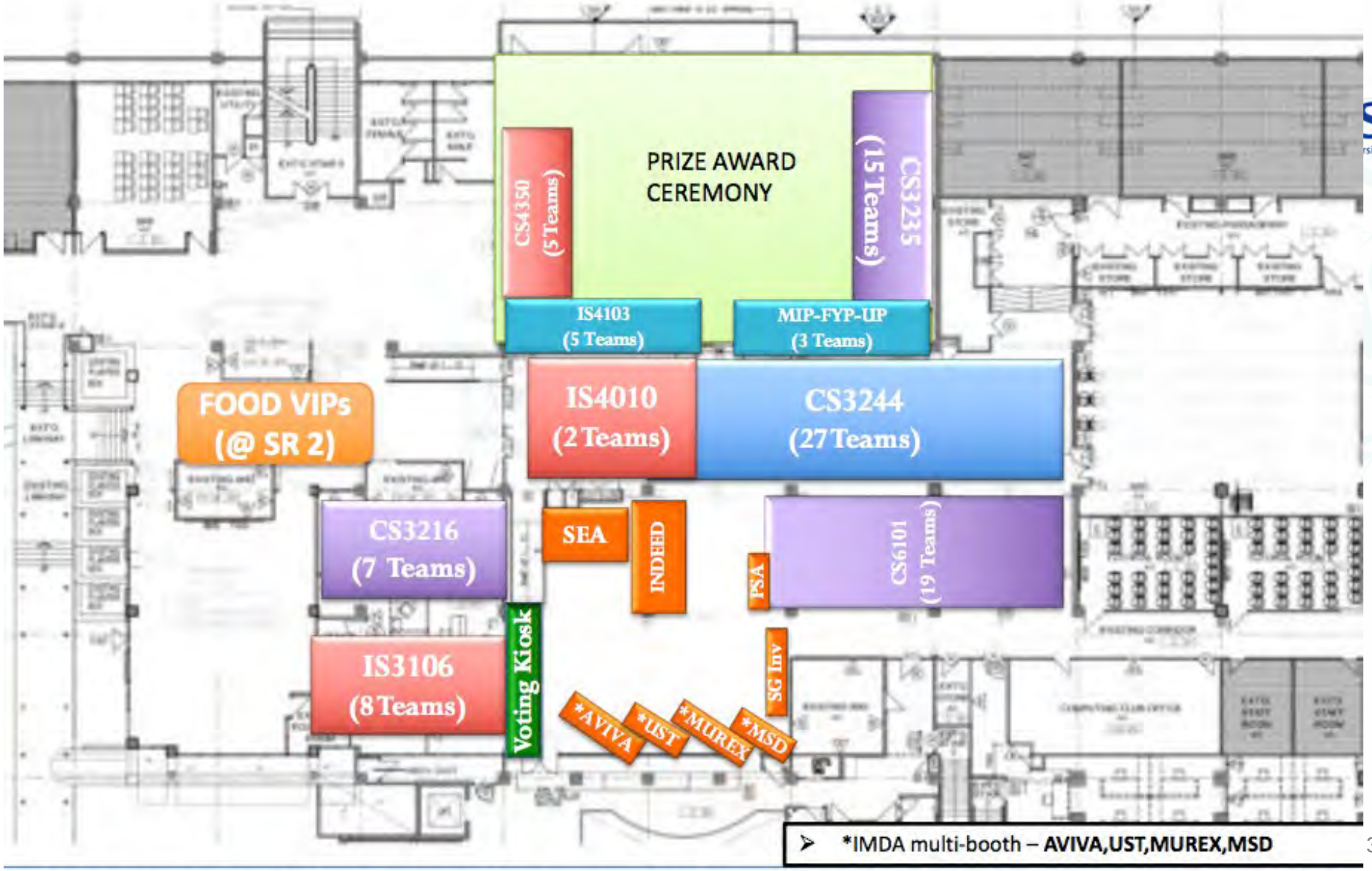
With the extended deadline, during the two hours of STePS presentation, your teams can review and grade your peers' poster and videos live.

You should also review the posters and videos beforehand instead if that fits your timing better.

A part of your team should stay at your poster to present to peers, while a part of your team circulates to peer grade other projects.

Show interest and civility to your peer's projects. Feel free to vote and comment on other projects that you find interesting, beyond the projects required of your peer grading.

- Peer and general audience voting will net winning teams prizes, but will not directly influence their project grades.



COM 1 (Level 1) – SR1, Cerebro, SR1 Foyer., Food for guests @ COM1 (Level B1)- B1 open area* (*not in map). OVERALL TOTAL : 91 projects.